

# Training Improvements (augmentation for fusion + noise)

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PAUT Explainable-AI Digital Twin — weld defect inspection from Total Focusing Method (TFM) images.

These changes target the weak points sir flagged (fusion, noise robustness) by enriching the **training data**, not the model. The SCN-Attention U-Net architecture and the fixed scattering filters are **unchanged** — so there is no risk of the collapse a redesign would cause. The improvements only take effect after a **retrain on GPU** (Colab/Kaggle); the current production checkpoint is untouched until then.

## What was added (`configs/model.yaml` → `augment`, **train split only**)

| Knob                               | Value      | Fixes                     | How it helps                                                                                                                                                                                                                                                                                         |
|------------------------------------|------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>noise_p / noise_sigma</code> | 0.3 / 0.04 | Noise robustness (#5)     | Adds additive Gaussian noise to training images so the seg model learns to keep finding defects under acquisition noise.                                                                                                                                                                             |
| <code>composite_p</code>           | 0.25       | Multi-defect fusion (#10) | With 25% probability, a training image is max-blended with one of the <b>other</b> class and their masks are unioned — so the model sees <b>both porosity and slag in one image</b> and learns to segment both. Every original image is single-type, which is exactly why fusion detection was weak. |

Both are train-only (never val/test) and gated by config (default off → no behaviour change unless enabled). Verified: with `composite_p=1.0` every sampled training mask contains both flaw classes.

## What this does NOT fix

- **Porosity Dice (~0.45):** the root cause is pseudo-labels (they shrink porosity to tiny cores).

The real lever is hand-labelling ~30–50 images. Composite/noise augmentation may help a little but is not the main fix.

- **Extreme-noise type accuracy:** a fundamental signal-to-noise limit (see the type-classifier

trade-off in the robustness commit) — not addressed by training augmentation.

## How to activate (one Colab/Kaggle GPU run)

```
# in the Colab notebook (notebooks/train_colab.ipynb), same as before – the new augmentation
# is read from configs/model.yaml automatically:
py -3.14 -m src.models.train --config configs/model.yaml
# then re-evaluate and refresh the experiment numbers:
py -3.14 -m src.models.evaluate --ckpt checkpoints/scn_attn_unet_best.pt --split test
py -3.14 -m scripts.multidefect_fusion --pairs 8      # expect both-detected to rise
```

```
py -3.14 -m scripts.robustness --limit 10          # expect the noise curve to fall less steeply
```

After the run, commit the new checkpoint + refreshed `model_metrics.json` and the experiment outputs. The numbers will then reflect the improved, incubated model.